

Final SKAE benchmark report

Executive summary

This report contains completed smoke-scale benchmark evidence plus an explicit list of omitted full runs. Results from non-official fixtures are pipeline checks only and are not scientific evidence for the requested PDEBench or Silverbox benchmarks.

Completed benchmark pipelines in this run: lorenz96, pdebench_swe, silverbox. Models/baselines recorded: dense_controlled_kae, dense_conv_kae, dense_kae, dmd, linear_arx, persistence, skae_controlled_k_ll, skae_conv_k_ll, skae_k_ll, truncated_svd_dmd.

Repository and implementation audit

See `reports/repository\_audit.md`. The main repository `SparseKM` maps to `GenericKM`; its built-in sparsity penalty is latent-activation L1, while this suite labels operator-L1 runs as `skae\_k\_ll` or related benchmark adaptations.

Experimental protocol

The checked-in config is `configs/skae\_benchmark\_suite.yaml`. It uses trajectory-level splits, train-only normalization, AdamW, gradient clipping, saved best/final checkpoints, raw per-trajectory metrics, and 2,000 bootstrap resamples for summary CIs. The profile is explicitly reduced for smoke execution.

Dataset and split details

Rows recorded: 368. Failures or omissions recorded: 2. Lorenz-96 data were generated by the RK4 generator in this suite. PDEBench and Silverbox official data were not available in this environment and are listed under limitations when fixtures were used.

Hyperparameter selection procedure

The smoke run evaluates the configured sparsity coefficients and latent dimensions directly. It does not complete the full requested grid or the 5% validation-error sparse operating-point rule; that remains encoded as future configuration work rather than a supported conclusion.

Main quantitative results

benchmark	condition	model	horizon	mean	std	ci95_low	ci95_high	n
lorenz96	D64_F8.0_obs1.0_noise0.0	dense_kae	1	0.990665	0.00425316	0.987658	0.993673	2
lorenz96	D64_F8.0_obs1.0_noise0.0	dense_kae	10	0.994909	0.000638904	0.994458	0.995361	2
lorenz96	D64_F8.0_obs1.0_noise0.0	dense_kae	5	0.990402	0.00325056	0.988103	0.9927	2
lorenz96	D64_F8.0_obs1.0_noise0.0	dmd	1	1.44835	2.00515	0.0304898	2.86621	2

	lorenz96		D64_F8.0_obs1.0_noise0.0		dmd		10		5.61323		7.77863		0.112914		11.1136		2	
	lorenz96		D64_F8.0_obs1.0_noise0.0		dmd		5		3.49695		4.85923		0.060951		6.93294		2	
	lorenz96		D64_F8.0_obs1.0_noise0.0		persistence		1		0.256295		0.00826124		0.250453		0.262136		2	
	lorenz96		D64_F8.0_obs1.0_noise0.0		persistence		10		1.1456		0.0179467		1.13291		1.15829		2	
	lorenz96		D64_F8.0_obs1.0_noise0.0		persistence		5		0.778474		0.00655425		0.77384		0.783109		2	
	lorenz96		D64_F8.0_obs1.0_noise0.0		skae_k_l1		1		0.990665		0.00425316		0.987658		0.993673		2	
	lorenz96		D64_F8.0_obs1.0_noise0.0		skae_k_l1		10		0.994909		0.000638875		0.994458		0.995361		2	
	lorenz96		D64_F8.0_obs1.0_noise0.0		skae_k_l1		5		0.990402		0.00325056		0.988103		0.9927		2	
	lorenz96		D64_F8.0_obs1.0_noise0.0		truncated_svd_dmd		1		0.650716		0.210365		0.501965		0.799466		2	
	lorenz96		D64_F8.0_obs1.0_noise0.0		truncated_svd_dmd		10		0.783378		0.242557		0.611864		0.954891		2	
	lorenz96		D64_F8.0_obs1.0_noise0.0		truncated_svd_dmd		5		0.68185		0.217693		0.527917		0.835782		2	
	pdebench_swe		clean_or_fixture		dense_conv_kae		1		1.01038		0		1.01038		1.01038		1	
	pdebench_swe		clean_or_fixture		dense_conv_kae		10		0.999974		0		0.999974		0.999974		1	
	pdebench_swe		clean_or_fixture		dense_conv_kae		5		1.00666		0		1.00666		1.00666		1	
	pdebench_swe		clean_or_fixture		skae_conv_k_l1		1		1.02427		0		1.02427		1.02427		1	
	pdebench_swe		clean_or_fixture		skae_conv_k_l1		10		1.01251		0		1.01251		1.01251		1	
	pdebench_swe		clean_or_fixture		skae_conv_k_l1		5		1.04657		0		1.04657		1.04657		1	
	silverbox		free_run		dense_controlled_kae		10		1.25901		0		1.25901		1.25901		1	
	silverbox		free_run		dense_controlled_kae		100		1.16108		0		1.16108		1.16108		1	
	silverbox		free_run		dense_controlled_kae		50		1.19453		0		1.19453		1.19453		1	
	silverbox		free_run		linear_arx		10		0.240693		0		0.240693		0.240693		1	
	silverbox		free_run		linear_arx		100		0.145798		0		0.145798		0.145798		1	
	silverbox		free_run		linear_arx		50		0.152729		0		0.152729		0.152729		1	
	silverbox		free_run		skae_controlled_k_l1		10		1.04711		0		1.04711		1.04711		1	
	silverbox		free_run		skae_controlled_k_l1		100		1.06465		0		1.06465		1.06465		1	
	silverbox		free_run		skae_controlled_k_l1		50		1.05371		0		1.05371		1.05371		1	

Sparsity-accuracy tradeoffs

	benchmark		model		horizon		sparsity_coefficient		mean		ci95_low		ci95_high		n	
	---		---		---		---		---		---		---		---	
	lorenz96		dense_kae		1		0		0.990665		0.987658		0.993673		2	
	lorenz96		dense_kae		10		0		0.994909		0.994458		0.995361		2	
	lorenz96		dense_kae		5		0		0.990402		0.988103		0.9927		2	
	lorenz96		dmd		1		0		1.44835		0.0304898		2.86621		2	
	lorenz96		dmd		10		0		5.61323		0.112914		11.1136		2	
	lorenz96		dmd		5		0		3.49695		0.060951		6.93294		2	
	lorenz96		persistence		1		0		0.256295		0.250453		0.262136		2	

lorenz96	persistence	10	0	1.1456	1.13291	1.15829	2	
lorenz96	persistence	5	0	0.778474	0.77384	0.783109	2	
lorenz96	skae_k_ll	1	0.0001	0.990665	0.987658	0.993673	2	
lorenz96	skae_k_ll	10	0.0001	0.994909	0.994458	0.995361	2	
lorenz96	skae_k_ll	5	0.0001	0.990402	0.988103	0.9927	2	
lorenz96	truncated_svd_dmd	1	0	0.650716	0.501965	0.799466	2	
lorenz96	truncated_svd_dmd	10	0	0.783378	0.611864	0.954891	2	
lorenz96	truncated_svd_dmd	5	0	0.68185	0.527917	0.835782	2	
pdebench_swe	dense_conv_kae	1	0	1.01038	1.01038	1.01038	1	
pdebench_swe	dense_conv_kae	10	0	0.999974	0.999974	0.999974	1	
pdebench_swe	dense_conv_kae	5	0	1.00666	1.00666	1.00666	1	
pdebench_swe	skae_conv_k_ll	1	0.0001	1.02427	1.02427	1.02427	1	
pdebench_swe	skae_conv_k_ll	10	0.0001	1.01251	1.01251	1.01251	1	
pdebench_swe	skae_conv_k_ll	5	0.0001	1.04657	1.04657	1.04657	1	
silverbox	dense_controlled_kae	10	0	1.25901	1.25901	1.25901	1	
silverbox	dense_controlled_kae	100	0	1.16108	1.16108	1.16108	1	
silverbox	dense_controlled_kae	50	0	1.19453	1.19453	1.19453	1	
silverbox	linear_arx	10	0	0.240693	0.240693	0.240693	1	
silverbox	linear_arx	100	0	0.145798	0.145798	0.145798	1	
silverbox	linear_arx	50	0	0.152729	0.152729	0.152729	1	
silverbox	skae_controlled_k_ll	10	0.0001	1.04711	1.04711	1.04711	1	
silverbox	skae_controlled_k_ll	100	0.0001	1.06465	1.06465	1.06465	1	
silverbox	skae_controlled_k_ll	50	0.0001	1.05371	1.05371	1.05371	1	

Long-horizon stability analysis

Long-horizon evidence is limited to the configured smoke horizons. Valid-prediction-time, spectral-radius, and instability-adjacent diagnostics are present in ``results/raw_metrics.parquet``; no broad stability claim is supported.

Noise and partial-observability analysis

The smoke config ran full-observation, clean Lorenz-96 only. The requested noise and partial-observation grids were not executed.

High-dimensional PDE results

The PDE path completed on an explicit synthetic fixture because the official PDEBench ``2D_rdb_NA_NA.h5`` file was absent. These rows validate code paths and output schemas only.

Real-data Silverbox results

The Silverbox path completed on an explicit synthetic fixture because the official ``nonlinear_benchmarks`` loader was unavailable. These rows are not evidence on real measured data.

Structural and spectral analysis

The suite records raw K L1, exact zeros, effective densities at relative thresholds, average active entries per coordinate, spectral radius, and eigenvalue moduli. Latent support is not claimed to identify physical causal structure.

Computational-cost analysis

benchmark	model	metric_name	mean	n
lorenz96	dense_kae	effective_density_1e3	0.0625	1
lorenz96	dense_kae	inference_time_seconds	0.000359717	1
lorenz96	dense_kae	spectral_radius	1.00152	1
lorenz96	skae_k_l1	effective_density_1e3	0.0625	1
lorenz96	skae_k_l1	inference_time_seconds	0.000326668	1
lorenz96	skae_k_l1	spectral_radius	1.00152	1
pdebench_swe	dense_conv_kae	effective_density_1e3	0.917969	1
pdebench_swe	dense_conv_kae	inference_time_seconds	0.00402149	1
pdebench_swe	dense_conv_kae	spectral_radius	1.04266	1
pdebench_swe	skae_conv_k_l1	effective_density_1e3	0.917969	1
pdebench_swe	skae_conv_k_l1	inference_time_seconds	0.00383978	1
pdebench_swe	skae_conv_k_l1	spectral_radius	1.02918	1
silverbox	dense_controlled_kae	effective_density_1e3	0.9375	1
silverbox	dense_controlled_kae	inference_time_seconds	0.00827914	1
silverbox	dense_controlled_kae	spectral_radius	1.0119	1
silverbox	skae_controlled_k_l1	effective_density_1e3	0.953125	1
silverbox	skae_controlled_k_l1	inference_time_seconds	0.00531376	1
silverbox	skae_controlled_k_l1	spectral_radius	1.02005	1

Limitations and failed runs

The following entries are machine-readable in `results/failures.json`:

```

```json
[
 {
 "benchmark": "silverbox",
 "reason": "Official nonlinear_benchmarks loader unavailable; ran explicit non-official smoke fixture.",
 "status": "official_data_omitted"
 },
 {
 "benchmark": "pdebench_swe",
 "reason": "Official file missing at data/pdebench/2D_rdb_NA_NA.h5; ran explicit non-official smoke fixture.",
 "status": "official_data_omitted"
 }
]
```

```

Reproduction commands

Use the commands in `reports/reproduction\_instructions.md`.

Direct answers to the central questions

- Predictive accuracy lost with sparsity: not answered beyond smoke-scale configured comparisons.
- Sparsity effects on stability, interpretability, sample efficiency, or robustness: not established.
- Long free-running accuracy: measured only at smoke horizons.
- Sparse structure stability across seeds: not established; only seed 0 completed.
- Scaling to high-dimensional states: not established; PDE fixture and Lorenz smoke are too small.
- Hidden Markov state and real input-output data: Silverbox official data were not run.
- Real measured input-output performance: not supported by this run.